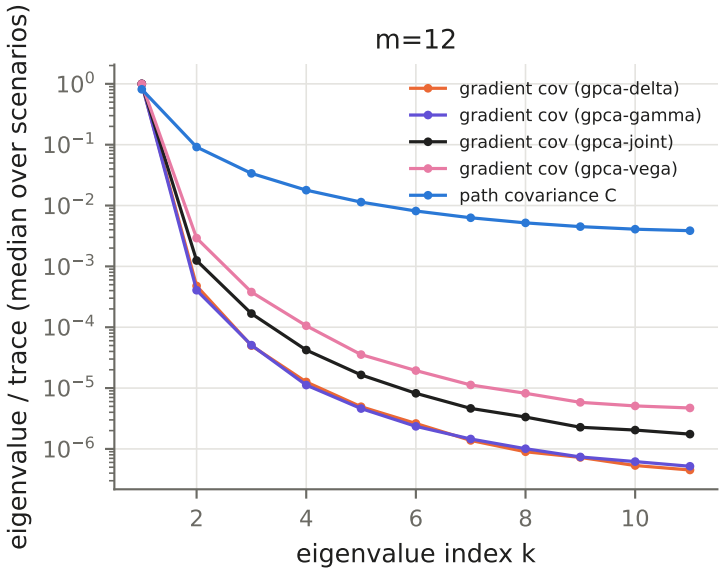Eigenvalue decay: path covariance vs Greek gradient covariances